

KON 313

Feedback Control Systems

Root Locus Techniques

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Introduction

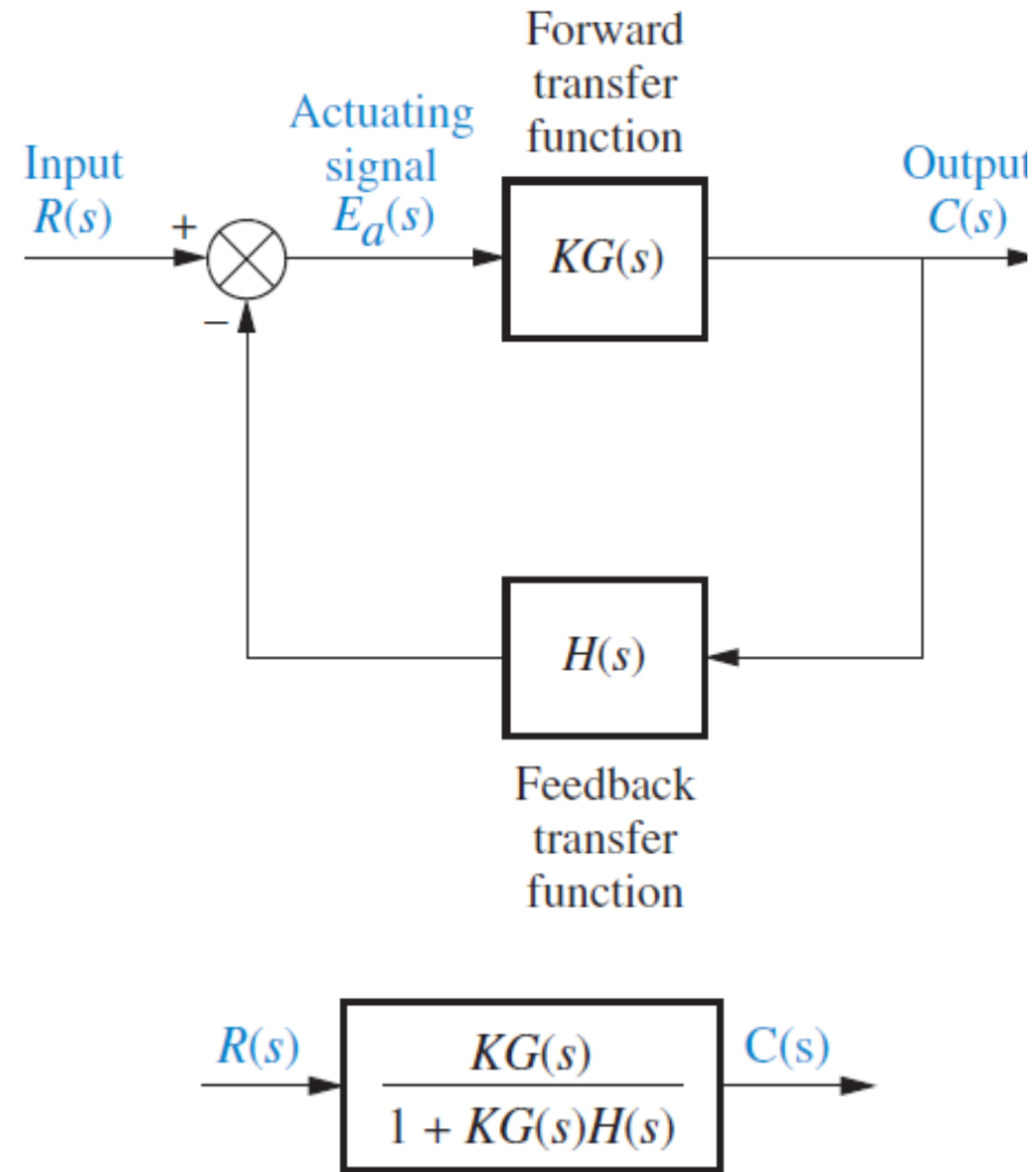
Recall. The basic characteristics of the **transient response** and **stability** of a closed-loop system is closely related to the location of the **closed-loop poles**.

- **Root locus** is a graphical presentation of the closed-loop poles as a system parameter is Varied. It, thus, is a powerful method of **analysis** and **design** for **stability** and **transient response**.
- The root locus is a graphical technique that gives us the **qualitative** description of a control system's performance and also serves as a powerful **quantitative** tool that yields more information than the methods already discussed.
- Up to this point, gains and other system parameters were designed to yield a desired transient response for only **first-** and **second-order** systems. Even though the root locus can be used to solve the same kind of problem, it has the ability to provide solutions for systems of **order higher than 2**.

Preliminaries: The Control System Problem

Consider a typical feedback closed-loop system.

- The **open-loop** transfer function is $KG(s)H(s)$.
 - We can **simply** determine the poles, since they arise from simple cascaded first- or second-order subsystems.
 - Variations in **K** **do not affect** the location of any pole of this function.
- The **closed-loop** transfer function is:
$$T(s) = KG(s)/[1 + KG(s)H(s)]$$
 - We **cannot** determine the poles of $T(s)$ unless we factor the denominator.
 - Also, the poles of $T(s)$ **change** with K .



Preliminaries: The Control System Problem

Consider a typical feedback closed-loop system.

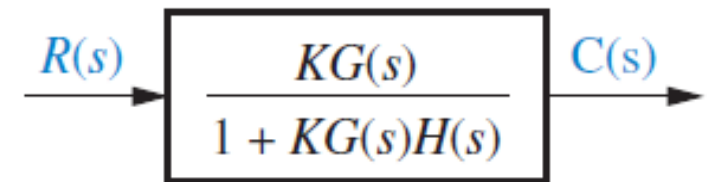
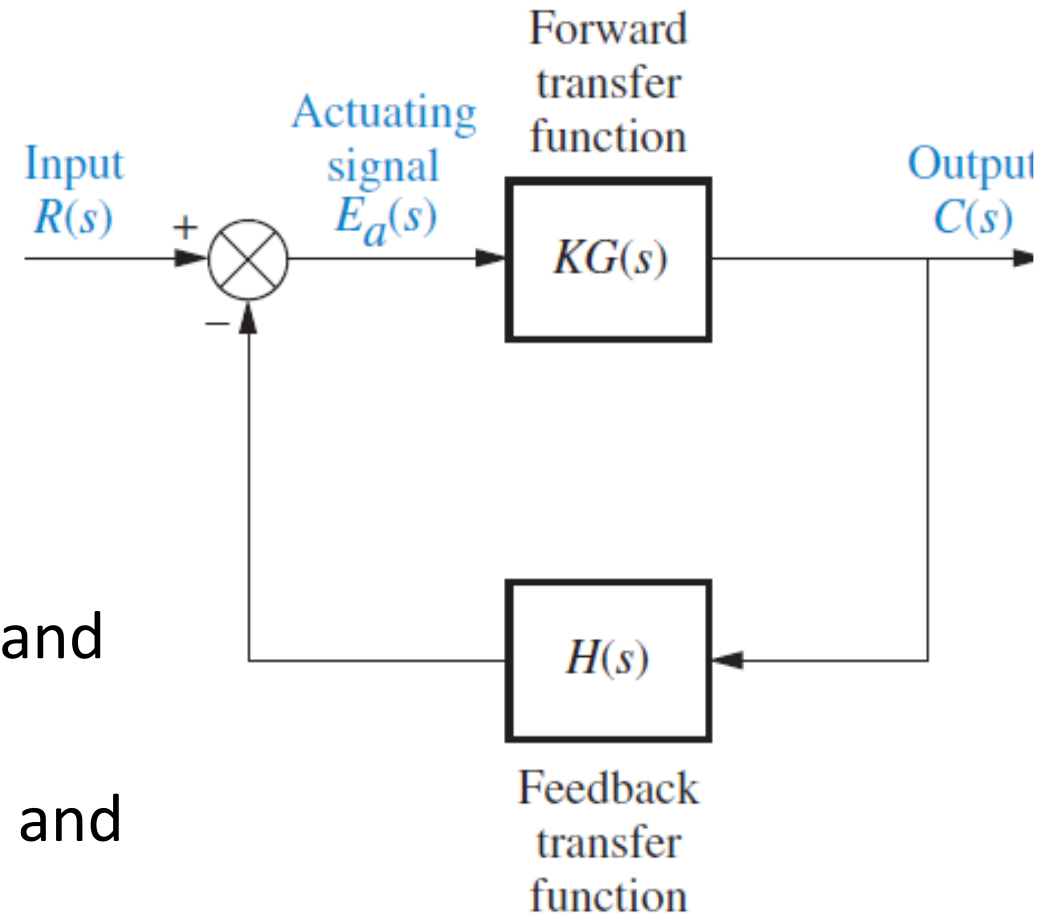
- Letting

$$G(s) = \frac{N_G(s)}{D_G(s)} \quad \text{an} \quad H(s) = \frac{N_H(s)}{D_H(s)}$$

Then

$$T(s) = \frac{KN_G(s)D_H(s)}{D_G(s)D_H(s) + KN_G(s)N_H(s)}$$

- Typically, we know the factors of the numerators and denominators of $G(s)$ and $H(s)$.
- Also, the zeros of $T(s)$ consist of the zeros of $G(s)$ and the poles of $H(s)$.
- The poles of $T(s)$ are not immediately known and in fact can change with K .



Preliminaries: Definitions

Characteristic Equation. The polynomial that forms the **denominator of the closed-loop** transfer function.

- Thus, the closed-loop poles are the roots of the characteristic equation.
- As the parameters of the open-loop transfer function varies, the characteristic equation changes. (The parameter is usually the gain, but **any other variable** of the open-loop transfer function may be used.)

Root-locus Method. A representation method in which the **roots** of the characteristic equation are plotted for all values of a **open-loop system parameter**.

- It is a powerful method of **analysis** and **design** for **stability** and **transient response**.
- Using this method the designer can predict the effects on the location of the closed-loop poles of varying the **gain** value or adding **open-loop poles** and/or **open-loop zeros**.
- The real power of Root-Locus Technique lies in its ability to provide solutions for systems of **order even higher than 2**.

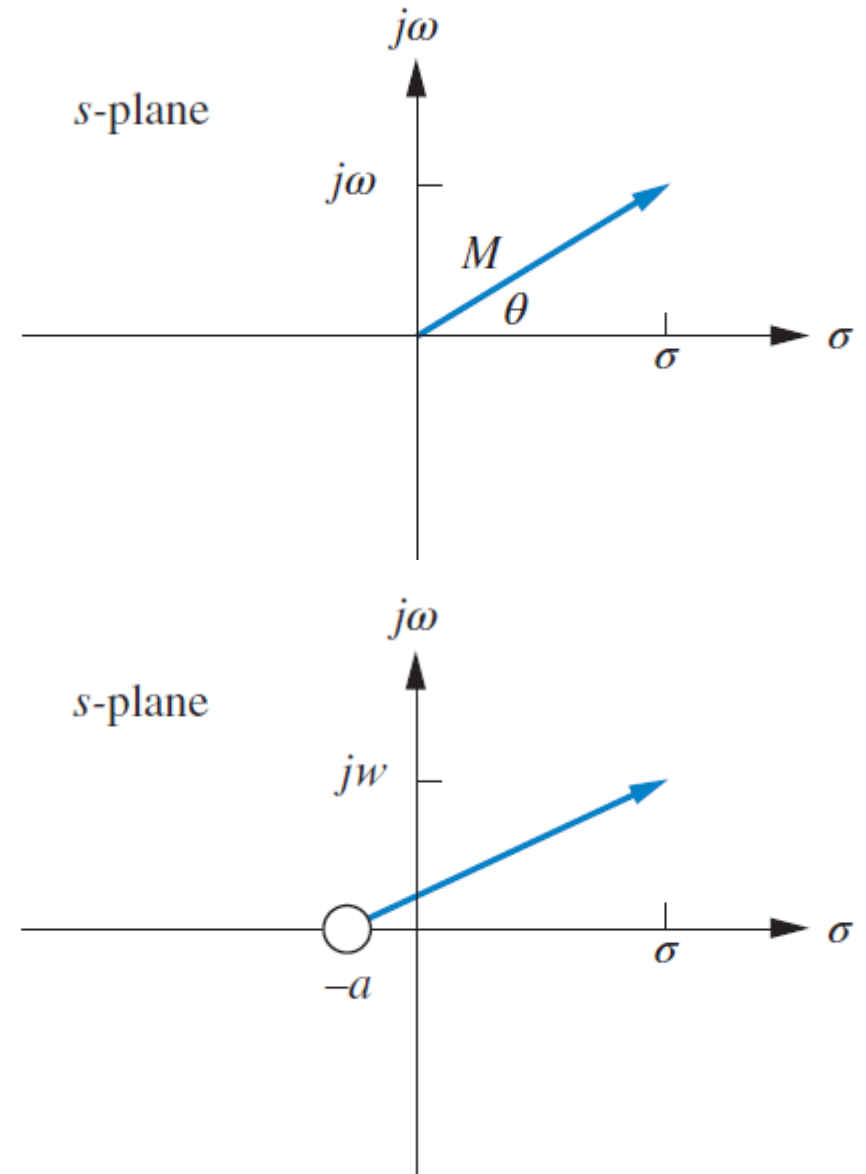
Preliminaries : Vector Representation of Complex Numbers

Any complex number can be

- described in Cartesian coordinates, $\sigma + j\omega$,
- graphically represented by a vector,
- also in polar form with magnitude M and angle θ , as $M\angle\theta$.

If the complex number, s , is substituted into a **complex function**, $F(s)$, another complex number will result.

Ex. $F(s) = (s + a)$ is a complex number and can be represented by a vector drawn from the zero of the function to the point s .



Preliminaries : Vector Representation of Complex Numbers

In a more general function:

$$F(s) = \frac{\prod_{i=1}^m (s + z_i)}{\prod_{j=1}^n (s + p_j)} = \frac{\text{numerator's complex factors}}{\text{denominator's complex factors}}$$

where the symbol Π means “product,” m = no. of zeros; and n = no. of poles.

The magnitude, M , and angle, θ , of $F(s)$ at any point, s , are

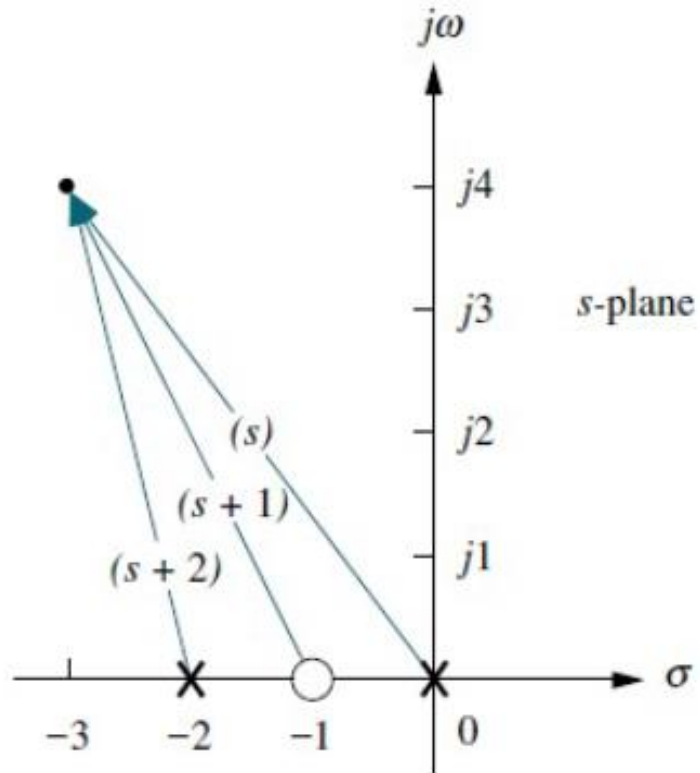
$$M = \frac{\text{numerator's lengths}}{\text{denominator's lengths}} = \frac{\prod_{i=1}^m |(s + z_i)|}{\prod_{j=1}^n |(s + p_j)|}$$

$$\begin{aligned}\theta &= \sum \text{zero angles} - \sum \text{pole angles} \\ &= \sum_{i=1}^m \angle(s + z_i) - \sum_{j=1}^n \angle(s + p_j)\end{aligned}$$

- A zero/pole **angle** is measured from the positive extension of the real axis, of a vector drawn from the zero/pole.

Preliminaries: Vector Representation of Complex Numbers

Ex. Find $F(s)$ at the point $s = -3+j4$ for $F(s) = \frac{s+1}{s(s+2)}$



The vector originating at the zero at -1 is

$$\sqrt{20} \angle 116.6^\circ$$

The vector originating at the pole at the origin is

$$5 \angle 126.9^\circ$$

The vector originating at the pole at -2 is

$$\sqrt{17} \angle 104.0^\circ$$

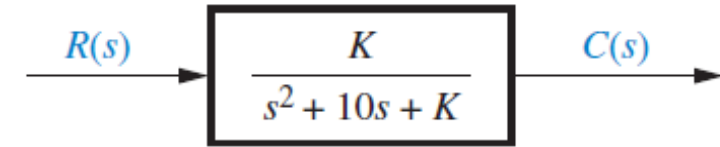
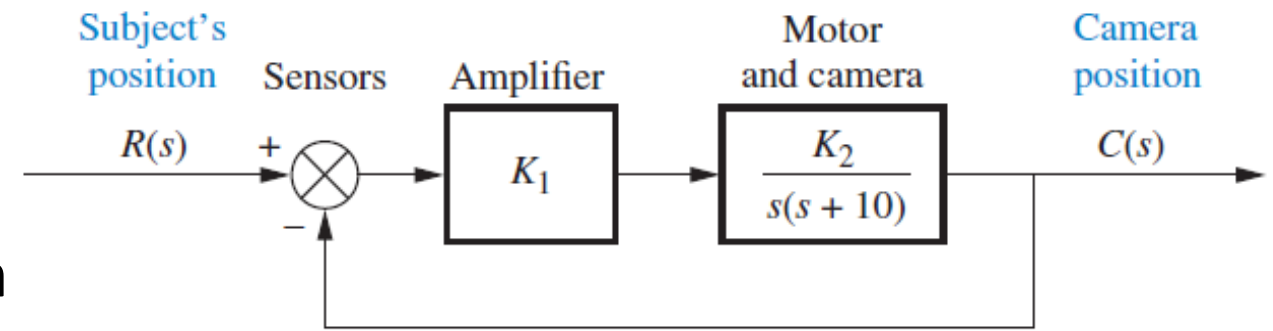
Substituting these values into the equations of the magnitude and the angle

$$M \angle \theta = \frac{\sqrt{20}}{5\sqrt{17}} \angle 116.6^\circ - 126.9^\circ - 104.0^\circ = 0.217 \angle -114.3^\circ$$

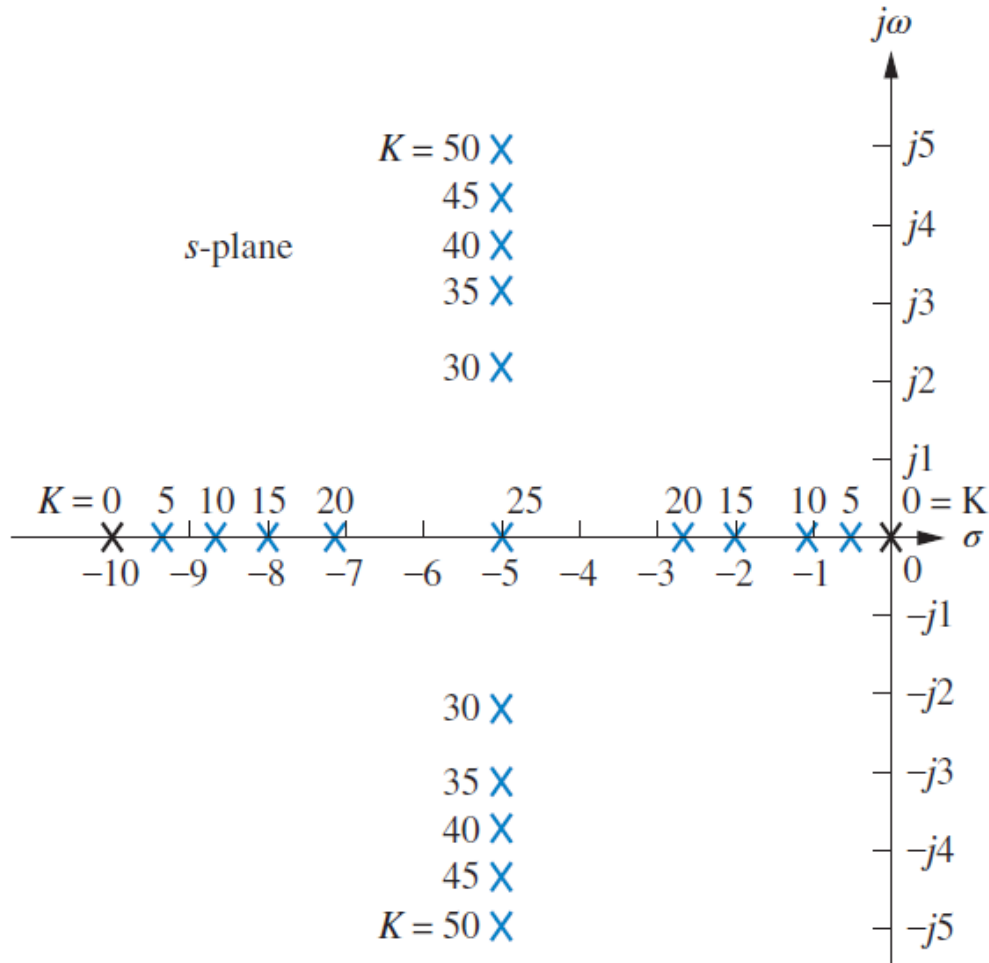
Defining the Root Locus

Ex. Consider the following system

The closed-loop poles of the system change location as the gain, K , is varied.



where $K = K_1 K_2$

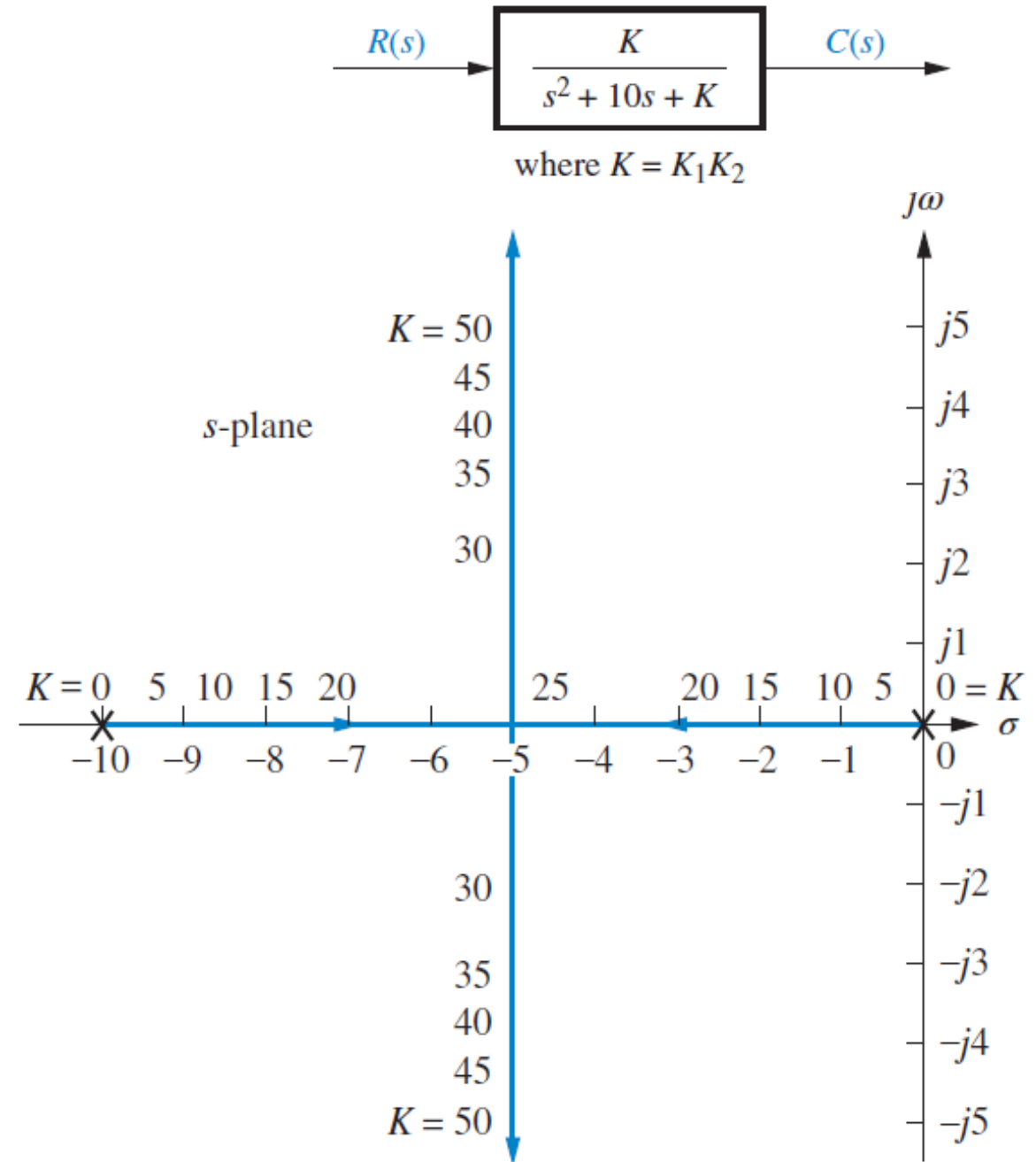


K	Pole 1	Pole 2
0	-10	0
5	-9.47	-0.53
10	-8.87	-1.13
15	-8.16	-1.84
20	-7.24	-2.76
25	-5	-5
30	$-5 + j2.24$	$-5 - j2.24$
35	$-5 + j3.16$	$-5 - j3.16$
40	$-5 + j3.87$	$-5 - j3.87$
45	$-5 + j4.47$	$-5 - j4.47$
50	$-5 + j5$	$-5 - j5$

Defining the Root Locus

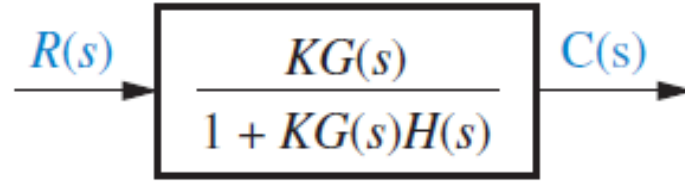
Ex. Cont.

- As the gain increases, the closed-loop pole, at -10 for $K = 0$, moves toward the right, and the closed-loop pole, at 0 for $K = 0$, moves toward the left. They meet at -5 , break away from the real axis, and move into the complex plane.
- This representation of the paths of the closed-loop poles as the gain is varied is called a **root locus**.
- For most of our work, the discussion will be limited to **positive gain**, or $K \geq 0$.



Properties of the Root Locus

Consider the transfer function of the typical closed-loop system, $T(s)$.



A value of s is a **closed-loop pole** when the characteristic polynomial becomes zero.

$$KG(s)H(s) = -1 = 1\angle(2k + 1)180^\circ \quad k = 0, \pm 1, \pm 2, \pm 3, \dots$$

Alternately,

$$|KG(s)H(s)| = 1$$

and

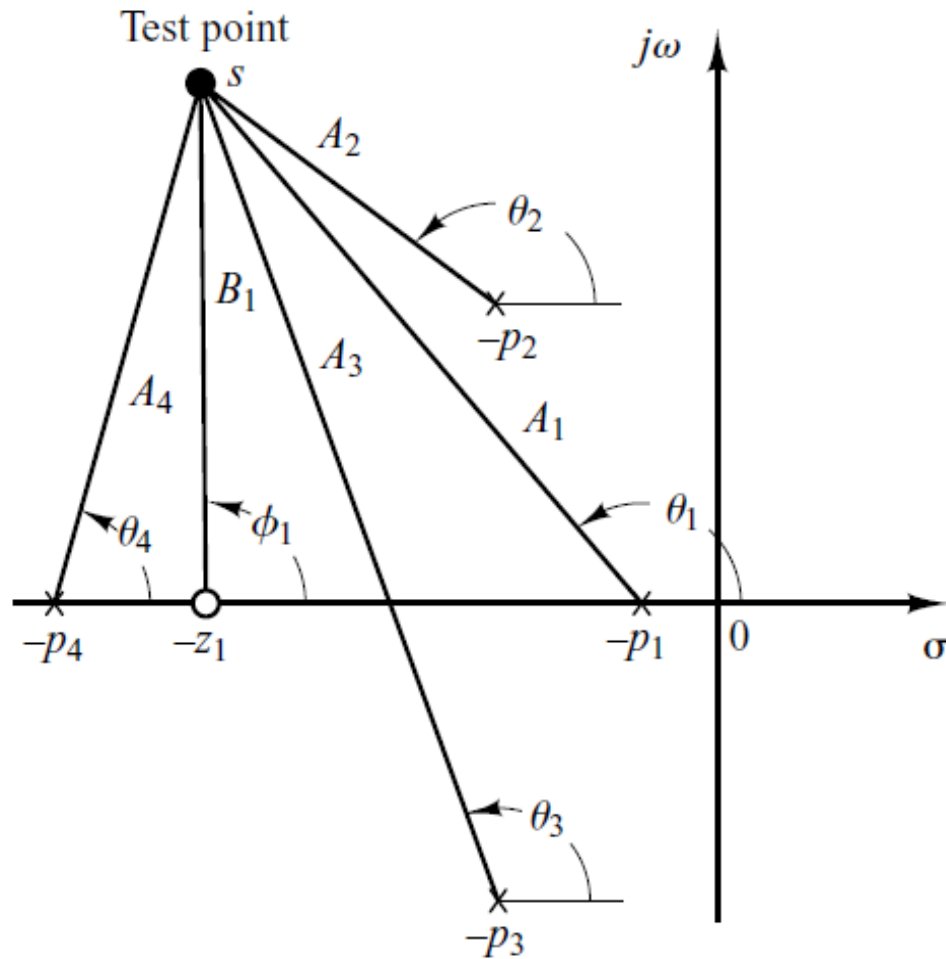
$$\angle KG(s)H(s) = (2k + 1)180^\circ$$

So, if substituting a value of s into the function $KG(s)H(s)$, the **angle** of the resulted complex number is an **odd multiple of 180°** , that value of s is a system **pole** for the value of K as:

$$K = \frac{1}{|G(s)||H(s)|}$$

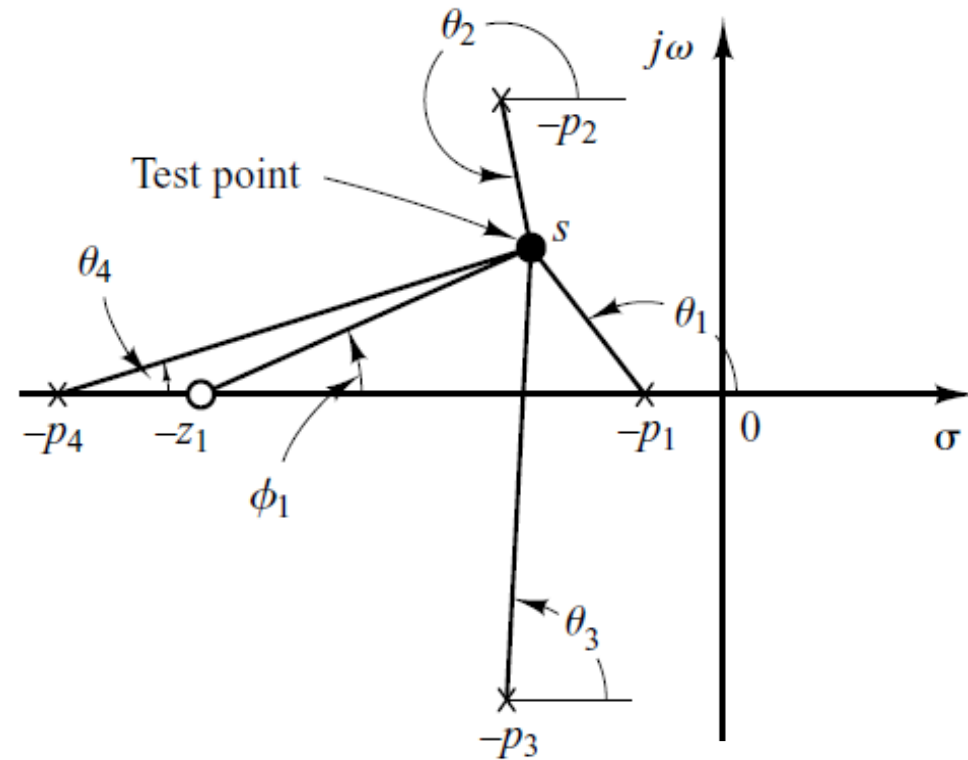
Properties of the Root Locus

Ex. to visualize the meaning of $|KG(s)H(s)| = 1$ and $\angle KG(s)H(s) = (2k + 1)180^\circ$



$$\angle G(s)H(s) = \phi_1 - \theta_1 - \theta_2 - \theta_3 - \theta_4$$

$$K G(s)H(s) = \frac{K(s + z_1)}{(s + p_1)(s + p_2)(s + p_3)(s + p_4)}$$



$$|G(s)H(s)| = \frac{KB_1}{A_1 A_2 A_3 A_4}$$

Sketching the Root Locus

The following rules allow us to sketch the root locus

Here, we assume that the parameter of interest is the gain K , where $K > 0$.

$$1 + \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} = 0$$

Rule 1. Number of branches. The number of branches equals the **number of closed-loop poles**.

- Each closed-loop pole moves as the gain is varied. Thus, each branch is the path of a pole.

Rule 2. Symmetry. The root locus is **symmetrical about the real axis**.

- If the roots of a polynomial with real coefficients are complex, they are definitely complex conjugate.

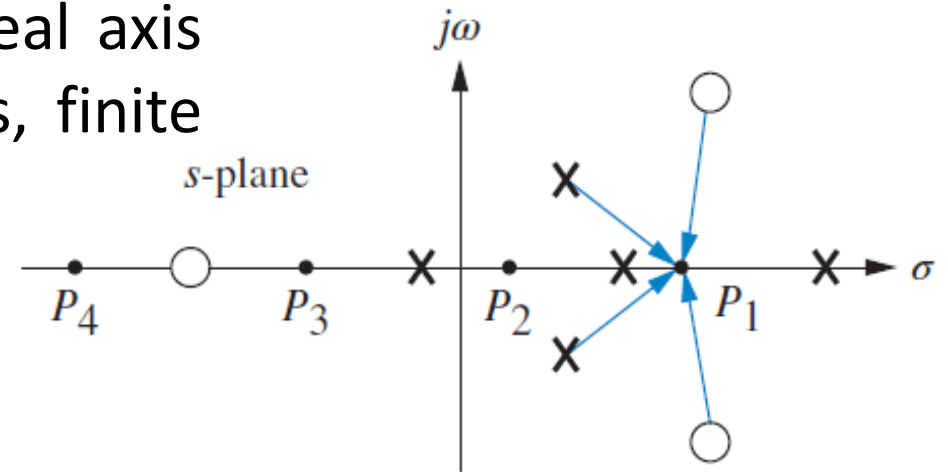
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Rule 3. Real-axis segments. Root locus on the real axis exists to the **left of an odd number** of real-axis, finite open-loop poles and/or finite open-loop zeros.



Rule 4. Starting and ending points. The root locus branches begin at the finite and infinite poles of $G(s)H(s)$ and end at the finite and infinite zeros of $G(s)H(s)$.

$$T(s) = \frac{KN_G(s)D_H(s)}{D_G(s)D_H(s) + KN_G(s)N_H(s)}$$

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$$1 + \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} = 0$$

Rule 5. The root locus approaches straight lines as asymptotes as the locus approaches infinity. The number of asymptotes equals $n - m$. The real-axis intersect and angle of the asymptotes is given by

$$\theta_a = \frac{(2k + 1)\pi}{\# \text{finite poles} - \# \text{finite zeros}}$$

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\# \text{finite poles} - \# \text{finite zeros}}$$

- A function can also have **infinite** poles and zeros. Every function of s has an equal number of poles and zeros if we include the infinite poles and zeros as well as the finite poles and zeros.

Sketching the Root Locus

The following rules allow us to sketch the root locus

Here, we assume that the parameter of interest is the gain K , where $K > 0$.

$$1 + \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} = 0$$

Rule 7. Real-Axis Break-in and Breakaway Points.

- If a root locus lies between two adjacent open-loop poles on the real axis, then there exists at least one breakaway point between the two poles.
- If the root locus lies between two adjacent zeros (one zero may be located at $-\infty$) on the real axis, then there always exists at least one break-in point between the two zeros.
- If the root locus lies between an open-loop pole and a zero (finite or infinite) on the real axis, then there may exist no breakaway or break-in points or there may exist both breakaway and break-in points.

Sketching the Root Locus

The following rules allow us to sketch the root locus

Here, we assume that the parameter of interest is the gain K , where $K > 0$.

$$1 + \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} = 0$$

Rule 7. Real-Axis Break-in and Breakaway Points. Are the real-valued solutions of the following equation

$$\frac{dK}{ds} = 0$$

The achieve **real-valued** s will be the considered point.

Proof: Break-in and breakaway points happens when K has its maximum or minimum value for the real values of s .

Sketching the Root Locus

The following rules allow us to sketch the root locus

Here, we assume that the parameter of interest is the gain K , where $K > 0$.

$$1 + \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} = 0$$

Rule 7. Real-Axis Break-in and Breakaway Points. The second method is called the transition method. Breakaway and break-in points satisfy the relationship

$$\sum_1^m \frac{1}{\sigma + z_i} = \sum_1^n \frac{1}{\sigma + p_i}$$

where z_i and p_i are the negative of the zero and pole values, respectively, of $G(s)H(s)$.

Sketching the Root Locus

The following rules allow us to sketch the root locus

Here, we assume that the parameter of interest is the gain K , where $K > 0$.

$$1 + \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} = 0$$

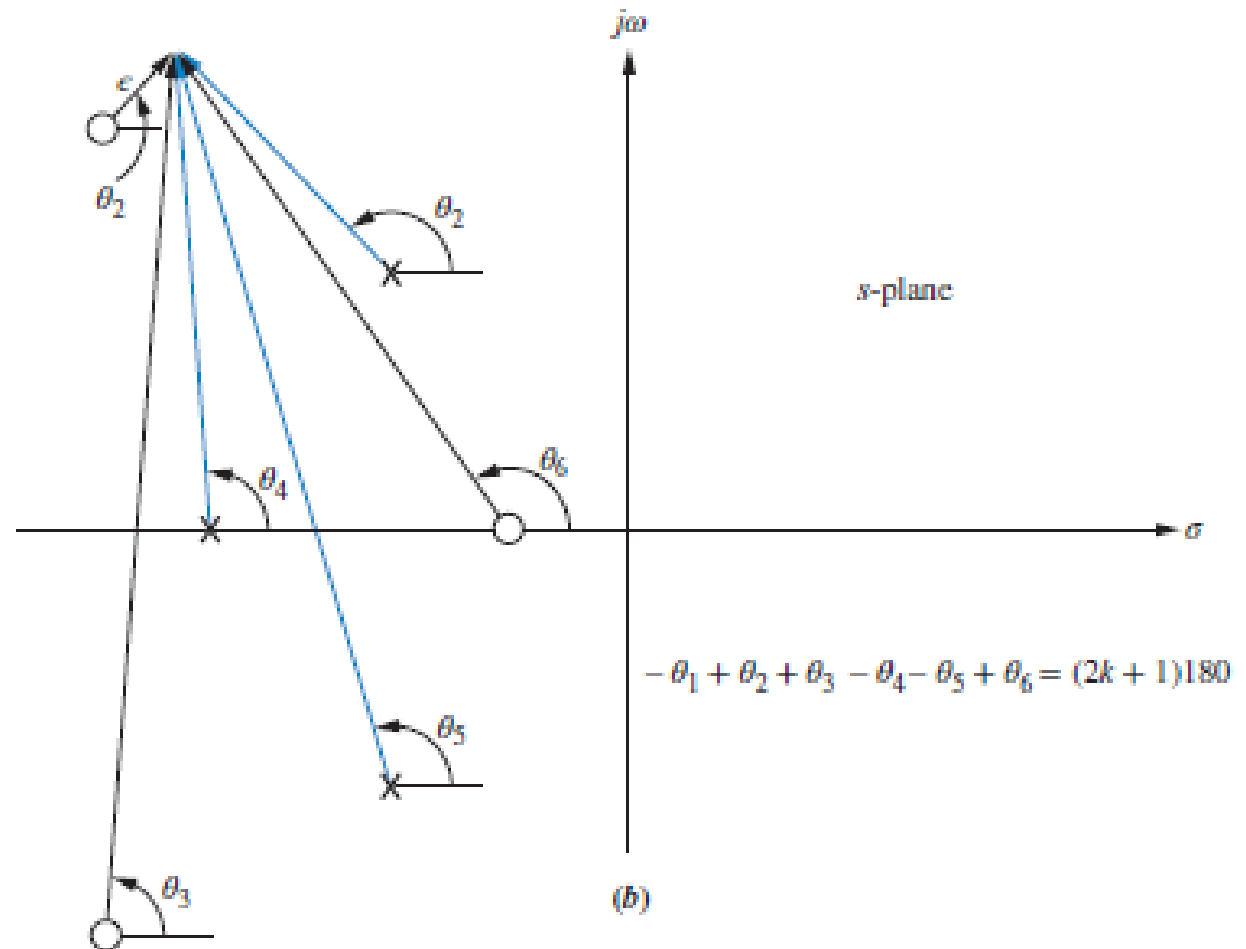
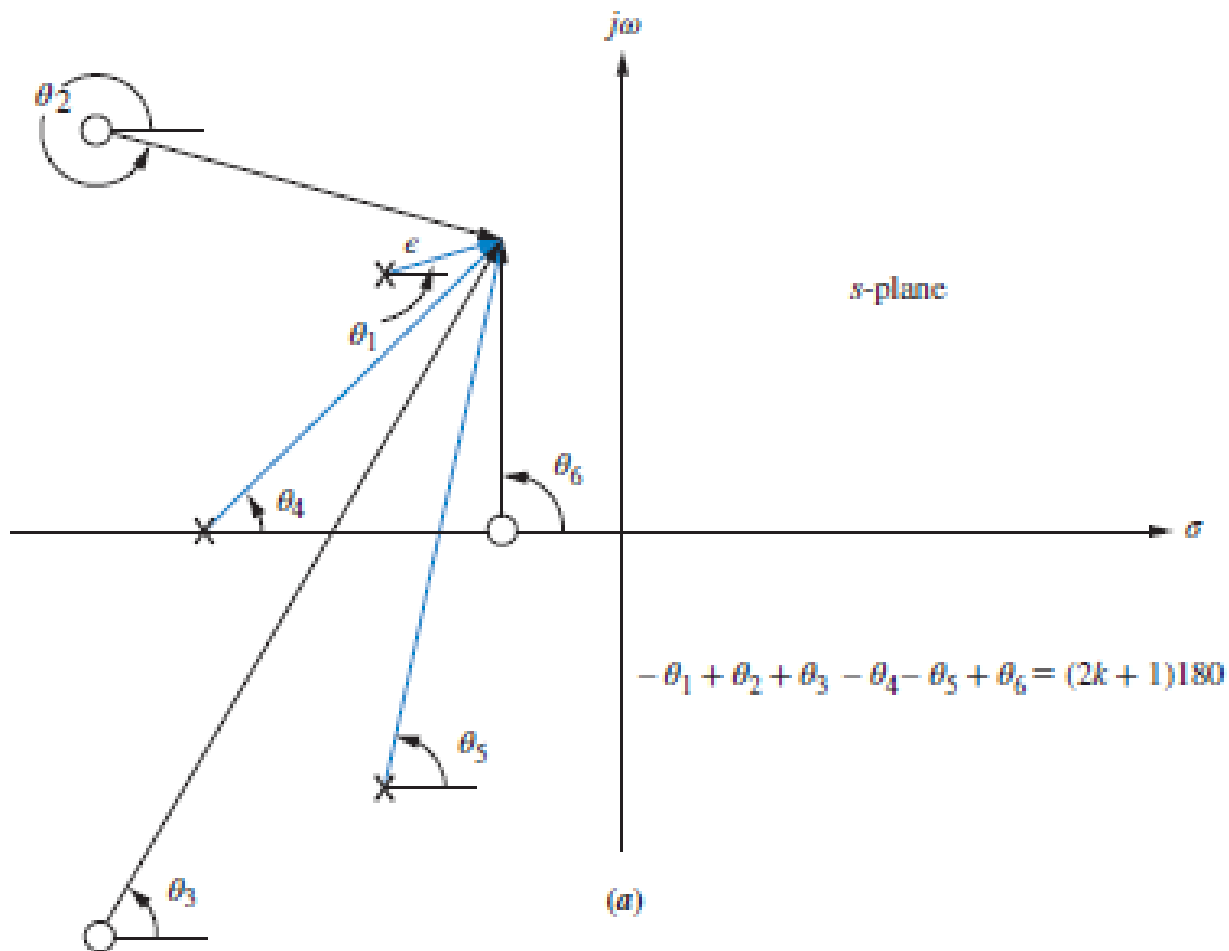
Rule 8. Angles of Departure and Arrival. We want to calculate the root locus departure angle from the complex poles and the arrival angle to the complex zeros.

Proof: If we assume a point on the root locus ϵ close to a complex pole, the sum of angles drawn from all finite poles and zeros to this point is an odd multiple of 180° . Except for the pole that is ϵ close to the point, we assume all angles drawn from all other poles and zeros are drawn directly to the pole that is near the point. Thus, the only unknown angle in the sum is the angle drawn from the pole that is ϵ close. We can solve for this unknown angle, which is also the angle of departure from this complex pole.

Sketching the Root Locus

The following rules allow us to sketch the root locus

Rule 8.



Sketching the Root Locus

The following rules allow us to sketch the root locus

Here, we assume that the parameter of interest is the gain K , where $K > 0$.

$$1 + \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} = 0$$

Rule 9. The $j\omega$ -Axis Crossings. We can use two methods:

1. The Routh-Hurwitz criterion,
 2. Put $s = j\omega$ in characteristic equations and make each real and imaginary part be equal to zero.
- The $j\omega$ -axis crossing is a point on the root locus that separates the stable operation of the system from the unstable operation.
 - The value of ω at the axis crossing yields the frequency of oscillation.

Sketching the Root Locus

The following rules allow us to sketch the root locus

Here, we assume that the parameter of interest is the gain K , where $K > 0$.

$$1 + \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)} = 0$$

Rule 10. Any point on root-locus is a closed-loop pole that is obtained at specific value of K . This K is calculated based on the equation.

$$|G(s)H(s)| = \frac{1}{K}$$

Rule 11. If the number of open-loop poles exceeds the number of finite zeros by three or more, there is a value of the gain K beyond which root loci enter the right-half s plane, and thus the system can become unstable

Sketching the Root Locus

A typical procedure for sketching the root-locus plot is as follows:

1. Find the characteristic equation and write it as

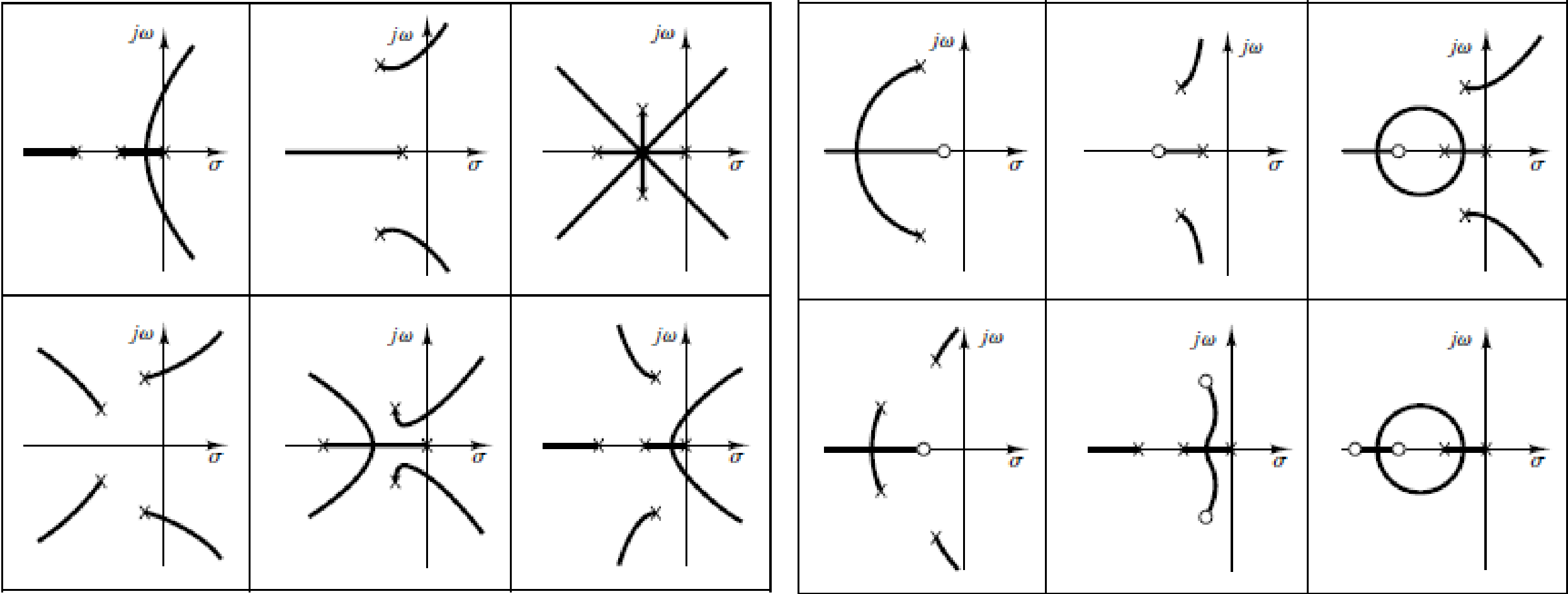
$$1 + K \frac{B(s)}{A(s)} = 0$$

2. Determine the location of open-loop poles and zeros.
3. Determine and plot the root-locus part on real-axis.
4. Plot the asymptotes.
5. Find the real-axis break-in and breakaway points (if there are any).
6. Find the departure/arrival angles from/to the complex poles/zeros (if there are any).
7. Find the $j\omega$ -Axis Crossing points (if there are any).

Sketching the Root Locus

Some Ex.

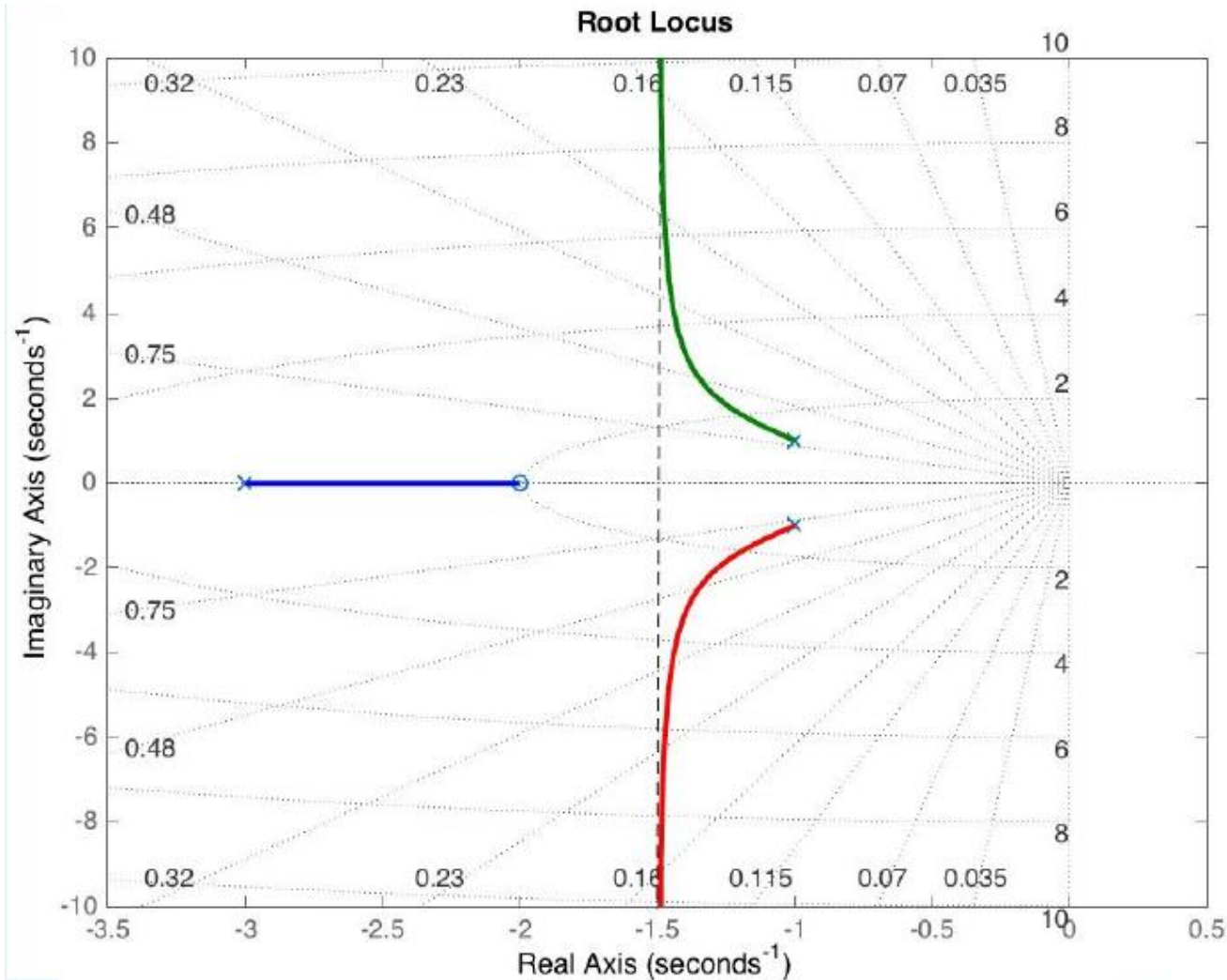
Table 6-1 Open-Loop Pole-Zero Configurations and the Corresponding Root Loci



Sketching the Root Locus

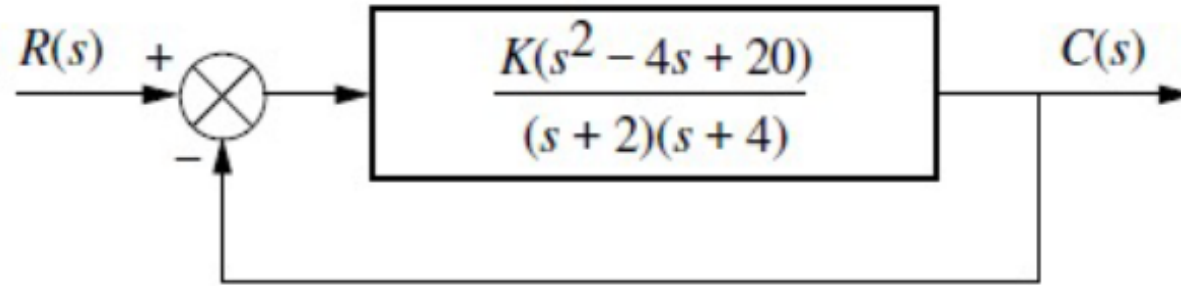
Ex.

$$G(s) = \frac{K(s + 2)}{(s + 3)(s^2 + 2s + 2)}$$

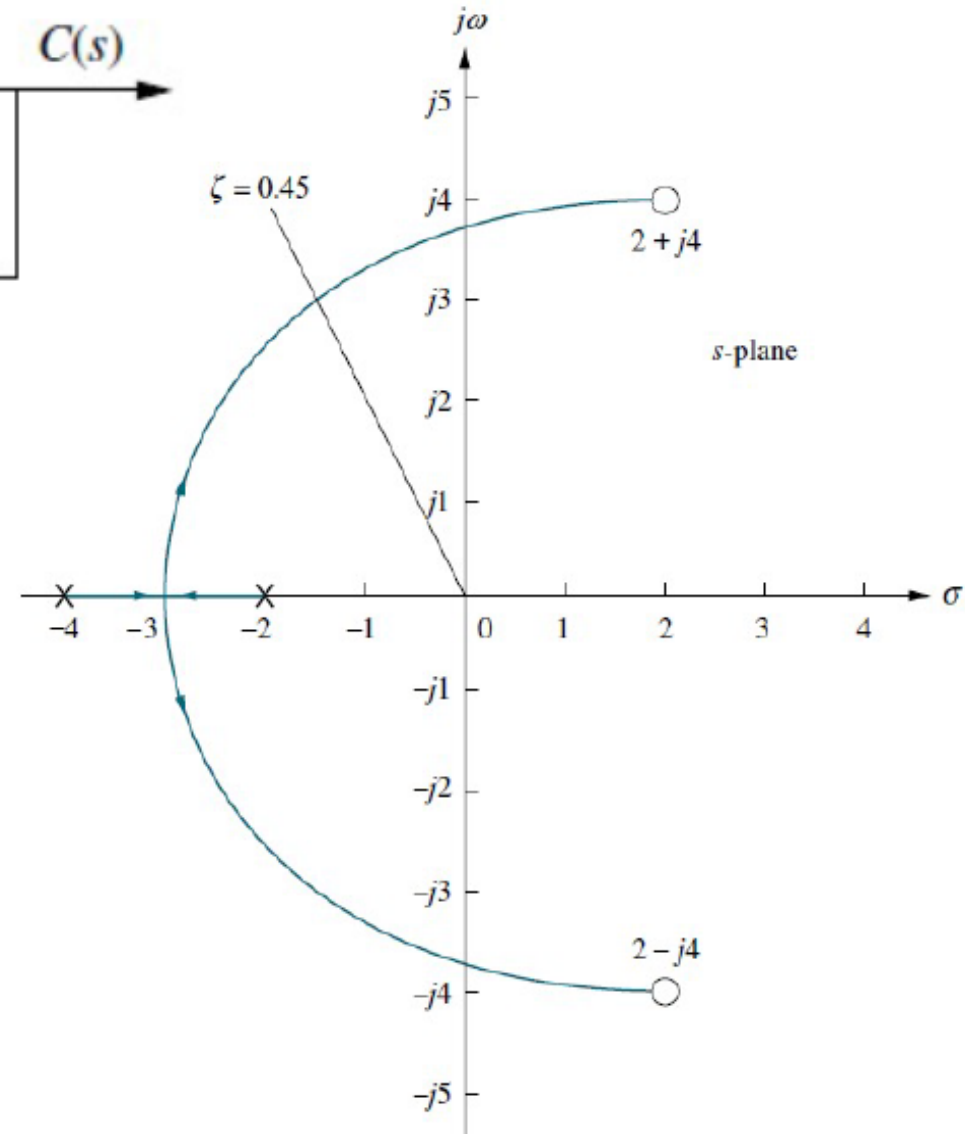


Sketching the Root Locus

Ex. Sketch the root locus for the system, then find the following.



- The exact point and gain where the locus crosses the 0.45 damping ratio line
- The exact point and gain where the locus crosses the $j\omega$ -axis
- The breakaway point on the real axis
- The range of K within which the system is stable

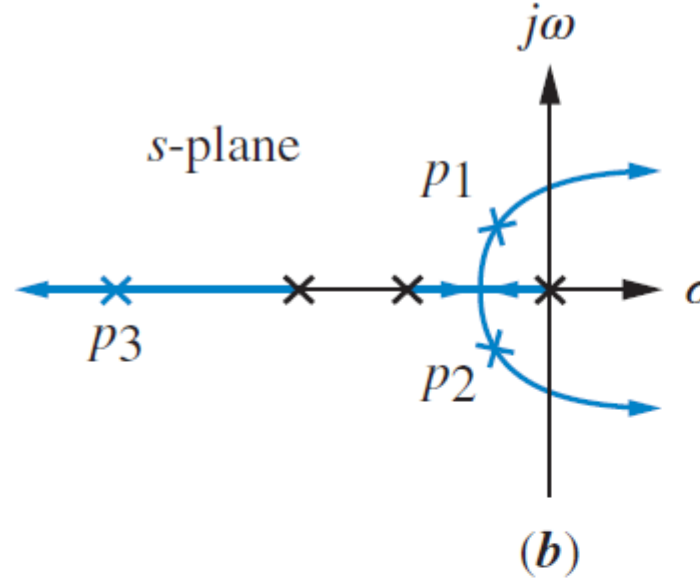
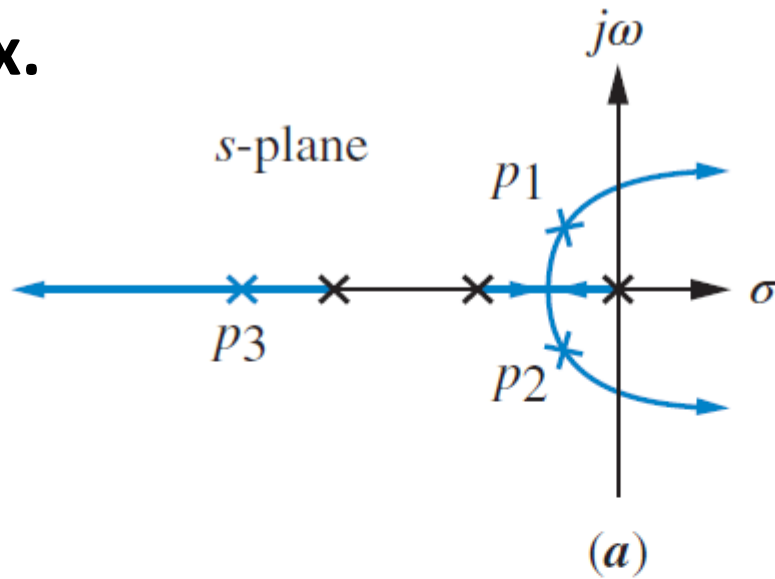


Transient Response Design via Gain Adjustment

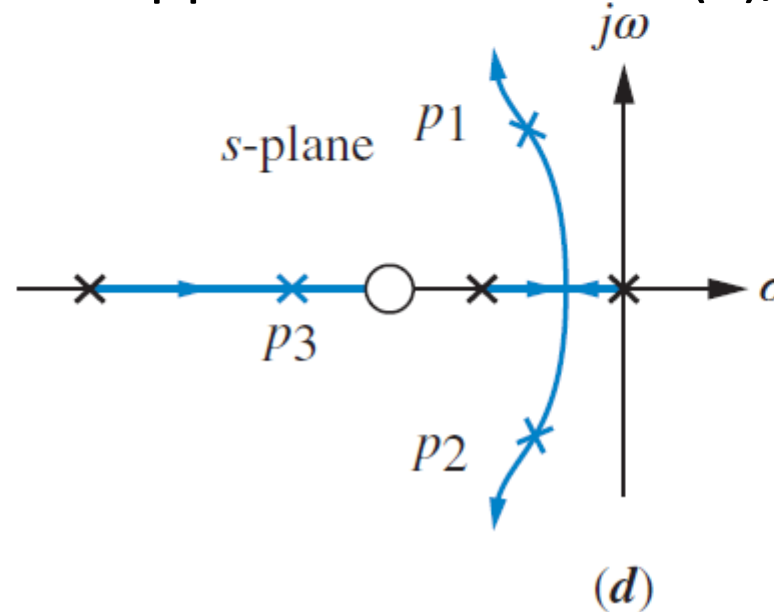
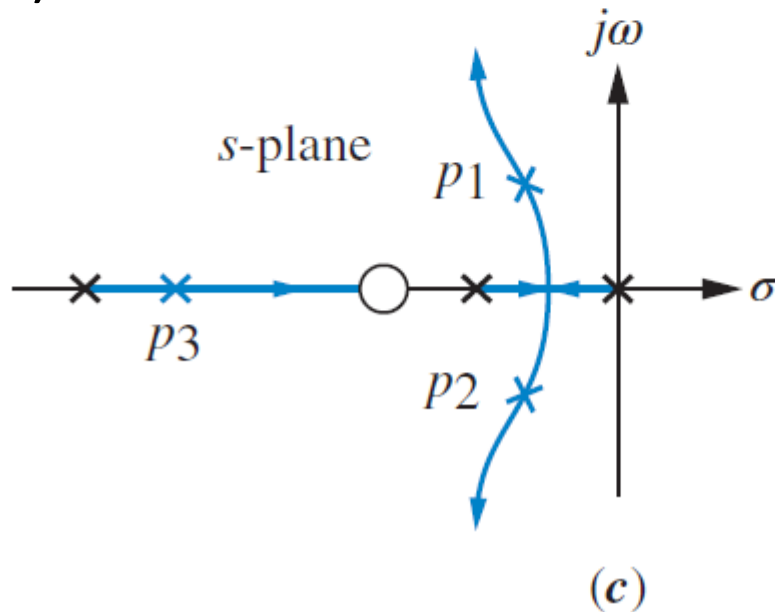
- It must be emphasized that the formulas describing percent overshoot, settling time, and peak time were derived only for a system with **two closed-loop complex poles and no closed-loop zeros**.
- The conditions justifying a second-order approximation are restated here:
 1. Higher-order **poles** are **much farther** into the left half of the s-plane than the dominant second-order pair of poles. The response that results from a higher-order pole does not appreciably change the transient response expected from the dominant second-order poles.
 2. Closed-loop **zeros** near the closed-loop second-order pole pair are nearly **cancelled** by the close proximity of higher-order closed-loop poles.
 3. Closed-loop **zeros** not cancelled by the close proximity of higher-order closed-loop poles are **far** removed from the closed-loop second-order pole pair.

Transient Response Design via Gain Adjustment

Ex.



(b)/(d) has a much better second-order approximation than (a)/(c)



- × Open-loop pole
- × Closed-loop pole
- Closed-loop zero

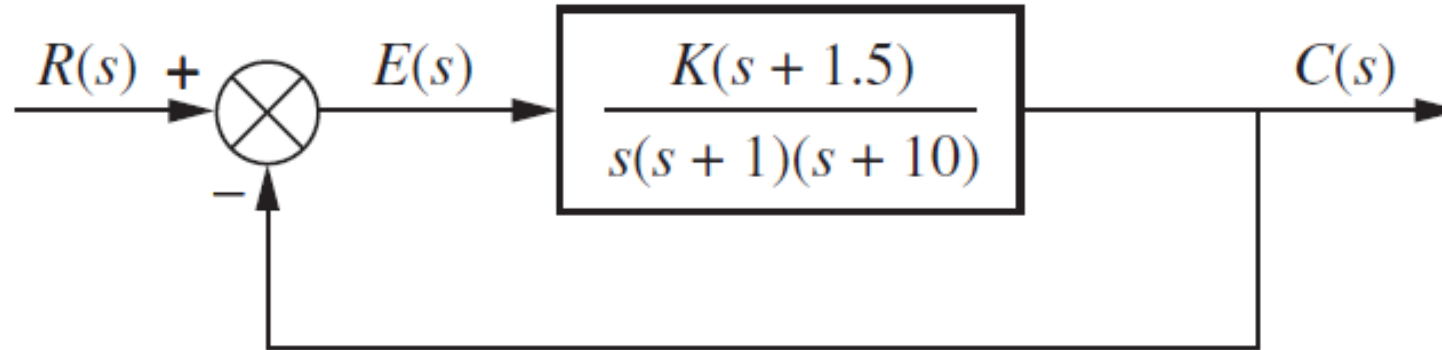
Transient Response Design via Gain Adjustment

Summarizing the **design** procedure for higher-order systems, we have:

1. Sketch the root locus for the given system.
2. Assume the system is a second-order system without any zeros and then find the gain to meet the transient response specification.
3. Justify your second-order assumption by finding the location of all **higher-order poles** and evaluating the fact that they are much farther (assumes a factor of five times farther) from the $j\omega$ -axis than the dominant second-order pair. Also, verify that **closed-loop zeros** are approximately **cancelled** by higher-order poles. If closed-loop zeros are not cancelled by higher-order closed-loop poles, be sure that the zero is **far** removed from the dominant second-order pole pair.
4. If the assumptions cannot be justified, your solution will have to be **simulated** in order to be sure it meets the transient response specification. It is a good idea to simulate all solutions, anyway.

Transient Response Design via Gain Adjustment

Ex. Consider the following system. Design the value of gain, K , to yield 1.52% overshoot. Also estimate the settling time, peak time, and steady-state error.

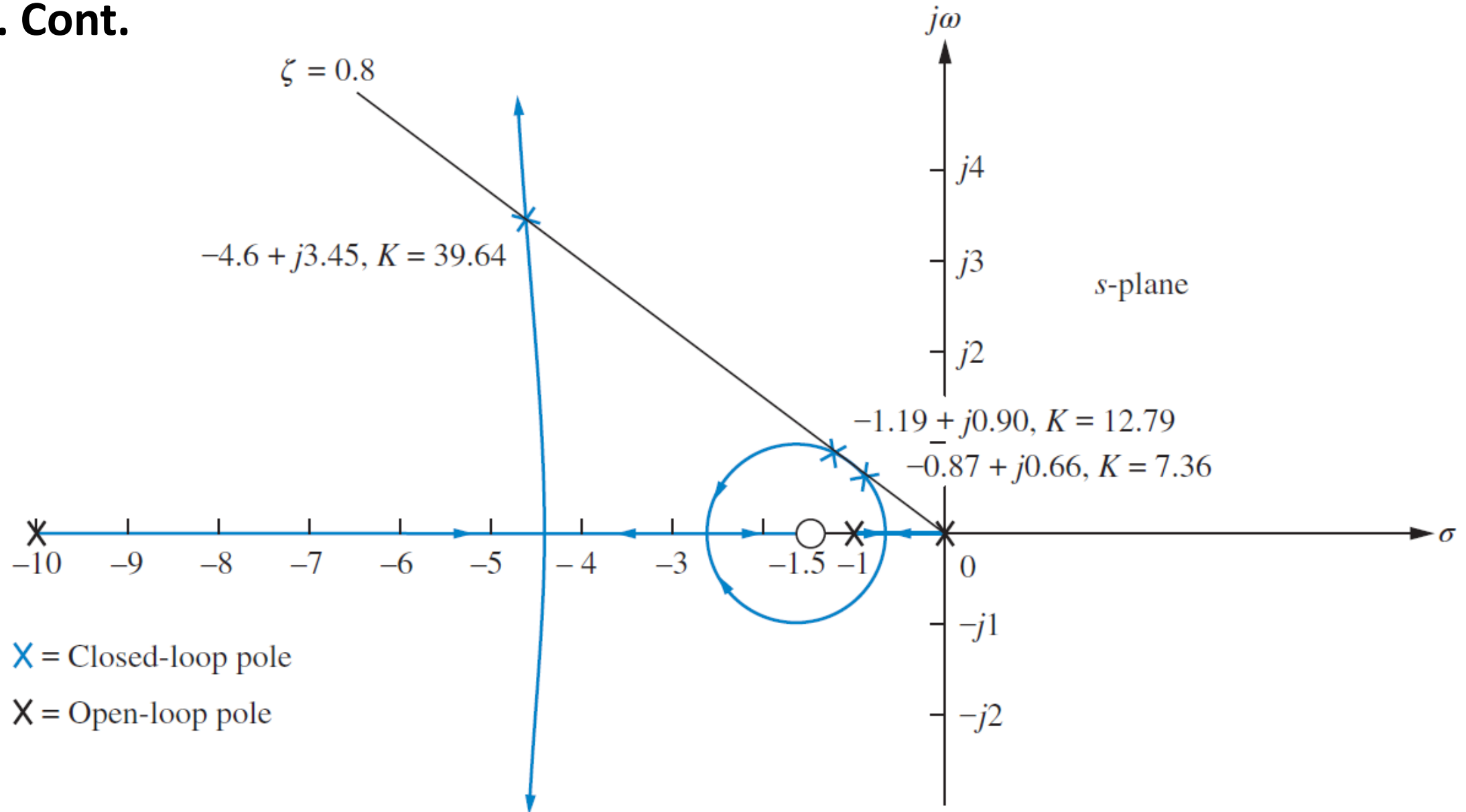


Soln.

- Assume that the system can be approximated by a second-order, under-damped system without any zeros.
- A 1.52% overshoot corresponds to a damping ratio of 0.8. Sketch this damping ratio line on the root locus.
- Use the root locus program to search along the 0.8 damping ratio line for the point where the angles from the open-loop poles and zeros add up to an odd multiple of 180° .

Transient Response Design via Gain Adjustment

Ex. Cont.



Transient Response Design via Gain Adjustment

Soln. Cont.

- Find the respective gain for each pair of poles.
- To test the assumption of a second-order system, calculate the location of the third pole, using the root locus program.
- Simulate or use partial-fraction expansion of the closed-loop step response.

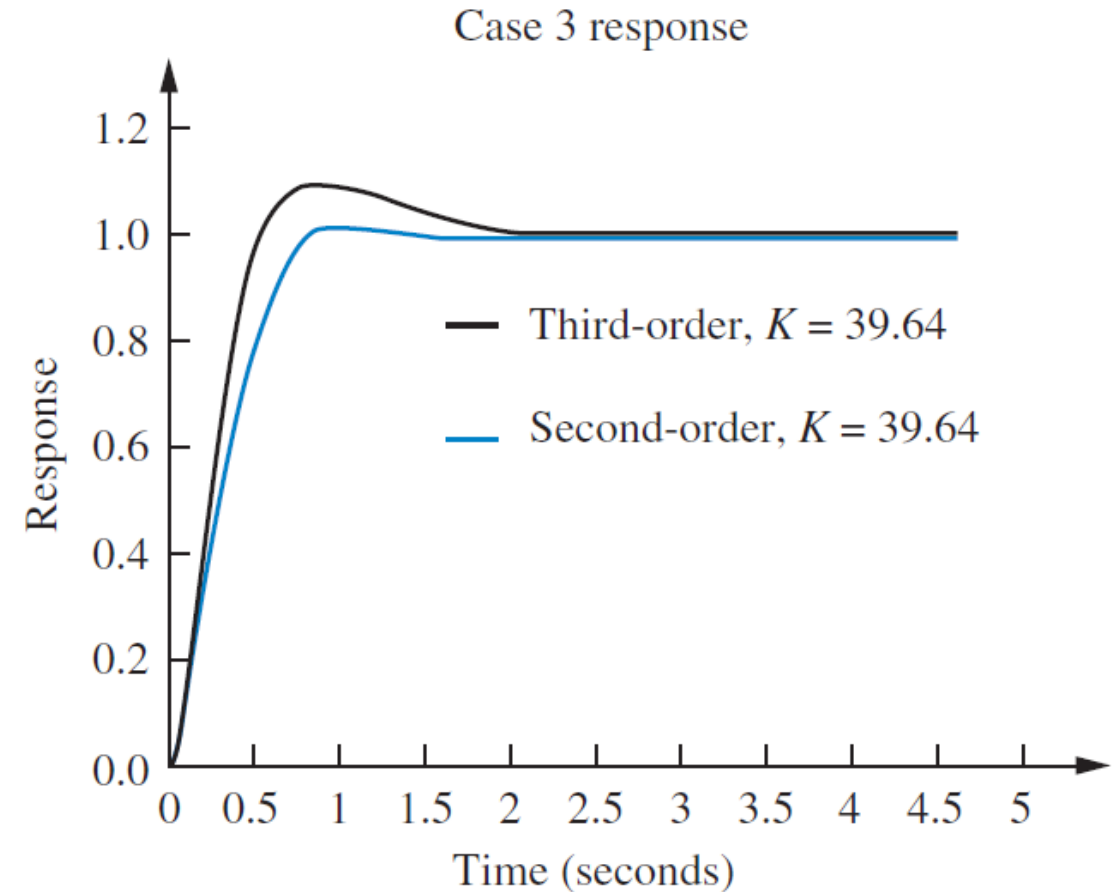
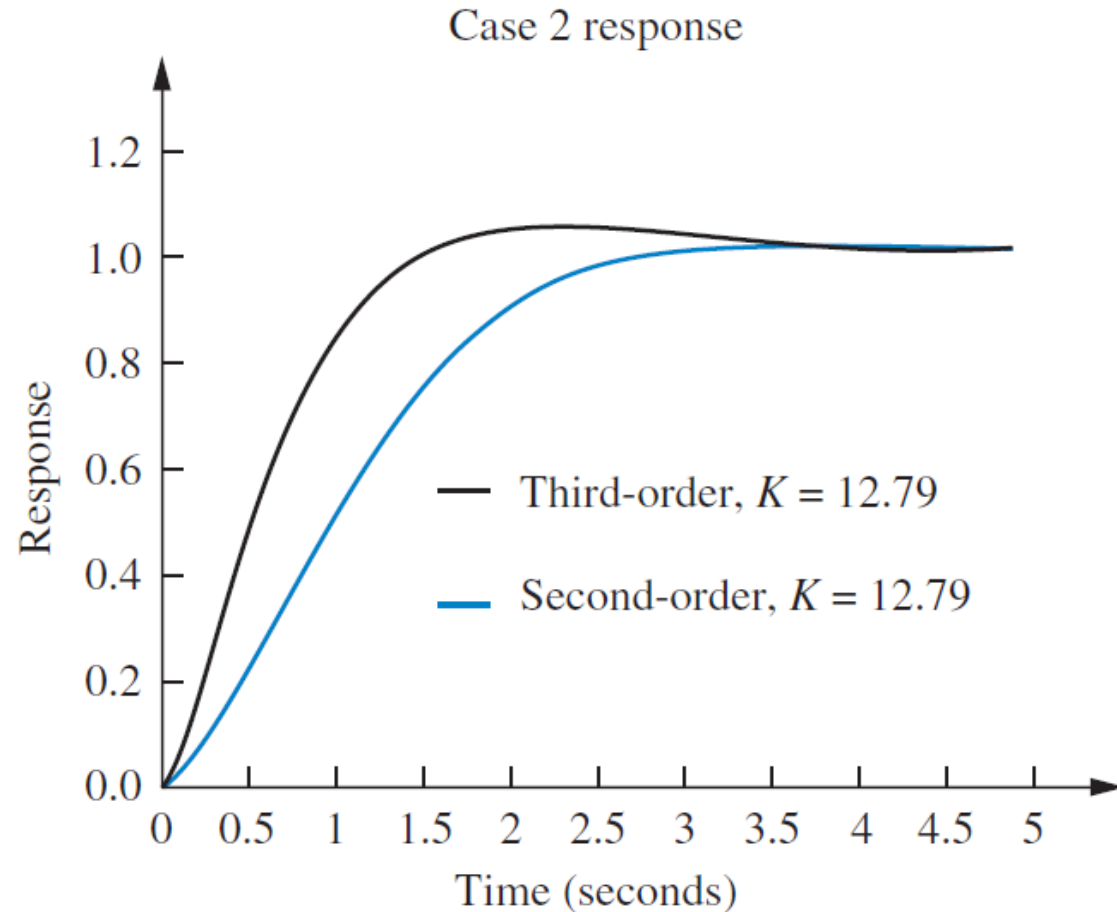
Result.

- Cases 1 and 2 yield third closed-loop poles that are relatively far from the closed-loop zero; There is no pole-zero cancellation, and a second-order system approximation is NOT valid.
- In Case 3, the third closed-loop pole and the closed-loop zero are relatively close to each other, and a second-order system approximation can be considered valid; There is a small difference in percent overshoot.

Transient Response Design via Gain Adjustment

Soln. Cont.

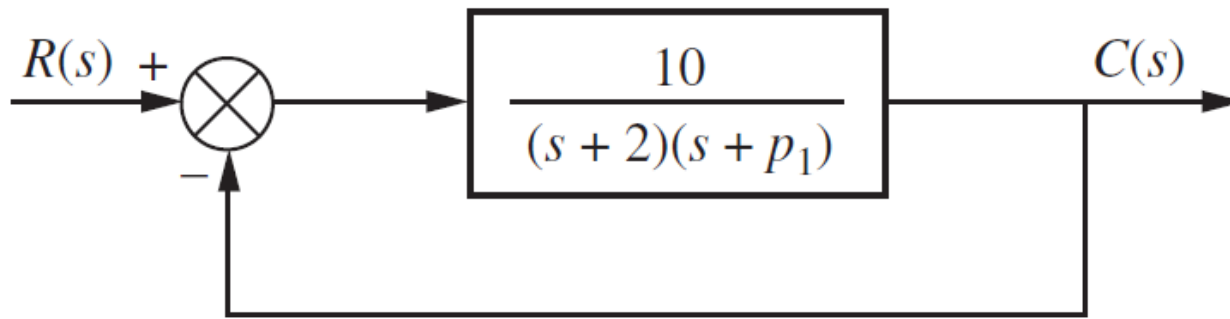
Result.



Generalized Root Locus

Up to this point we have always drawn the root locus as a function of the forward-path gain, K . The control system designer must often know how the closed-loop poles change **as a function of another parameter**.

Ex.



$$KG(s)H(s) = \frac{10}{(s + 2)(s + p_1)}$$

The problem is that p_1 is not a multiplying factor of the function, as the gain, K .

The solution is to create an equivalent system where p_1 appears as the forward-path gain.

$$T(s) = \frac{KG(s)}{1 + KG(s)H(s)} = \frac{10}{s^2 + (p_1 + 2)s + 2p_1 + 10}$$

Generalized Root Locus

Ex. Cont.

$$T(s) = \frac{KG(s)}{1 + KG(s)H(s)} = \frac{10}{s^2 + (p_1 + 2)s + 2p_1 + 10}$$

Isolating p_1

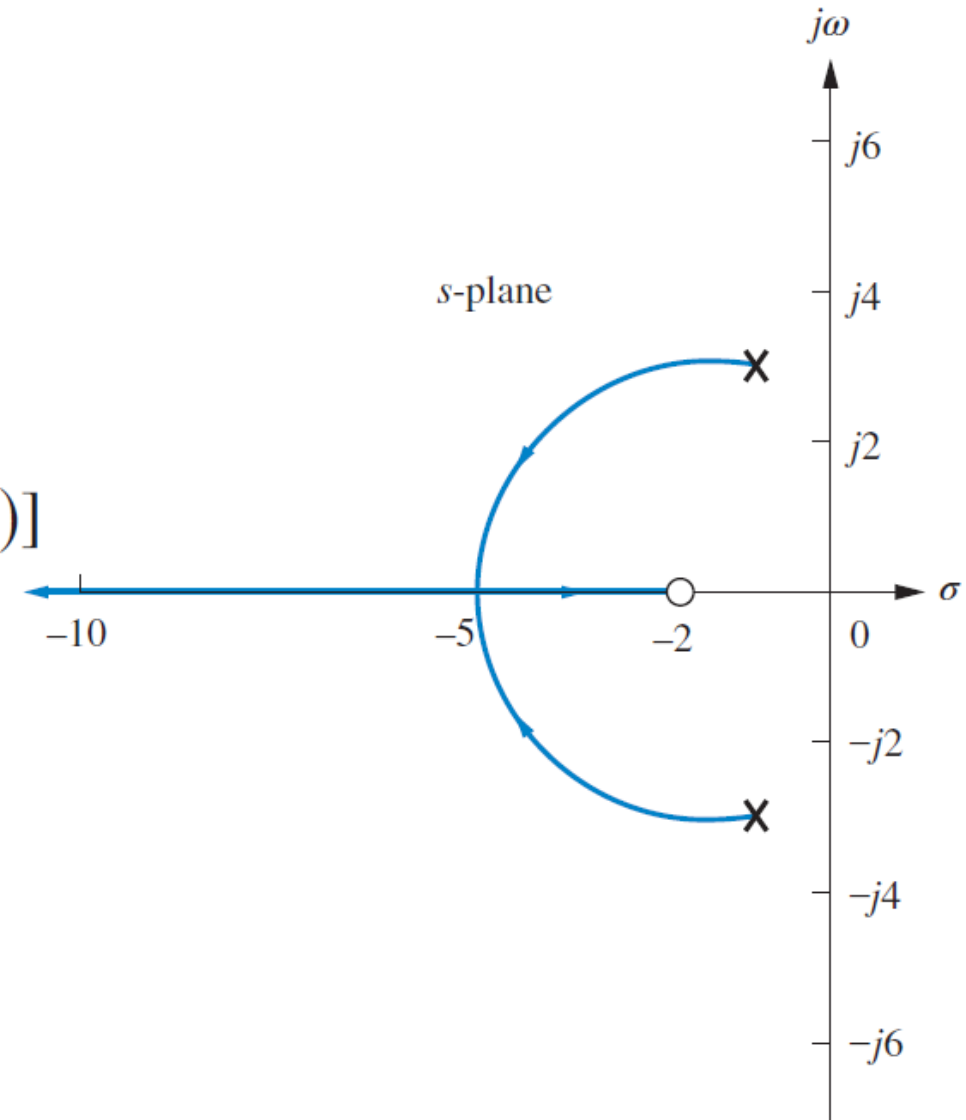
$$T(s) = \frac{10}{s^2 + 2s + 10 + p_1(s + 2)}$$

Convert the denominator to the form $[1 + p_1 G(s)H(s)]$

By dividing numerator and denominator by the term not included with p_1 .

$$T(s) = \frac{\frac{10}{s^2 + 2s + 10}}{1 + \frac{p_1(s + 2)}{s^2 + 2s + 10}}$$

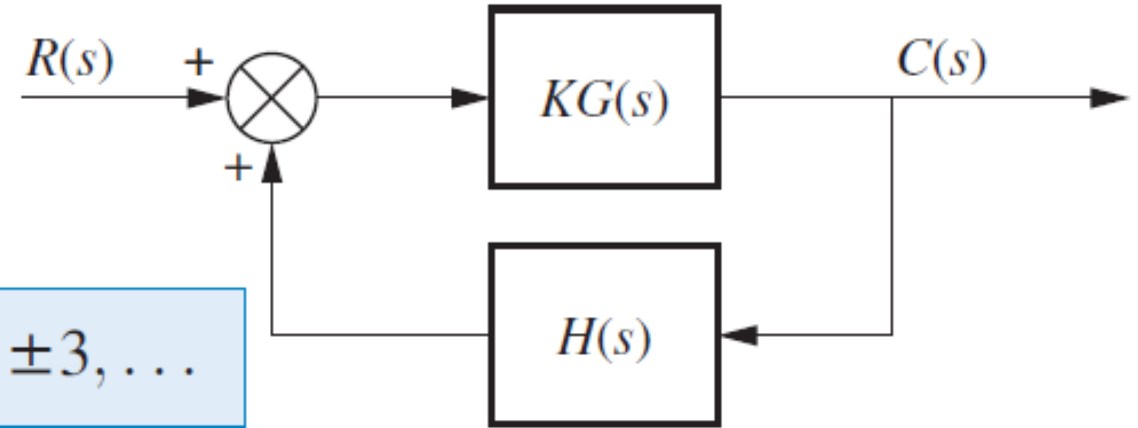
$$KG(s)H(s) = \frac{p_1(s + 2)}{s^2 + 2s + 10}$$



Root Locus for Positive-Feedback Systems

$$T(s) = \frac{KG(s)}{1 - KG(s)H(s)}$$

$$KG(s)H(s) = 1 = 1 \angle k360^\circ \quad k = 0, \pm 1, \pm 2, \pm 3, \dots$$



Rule 3. *Real-axis segments*: On the real axis, the root locus for positive-feedback systems exists to the left of an **even** number of real-axis, finite open-loop poles and/or finite open-loop zeros.

Rule 5. *Behaviour at infinity*.

$$\theta_a = \frac{k2\pi}{\# \text{ finite poles} - \# \text{ finite zeros}}$$